

HiggsCP Analysis

DOCUMENTATION AND GUIDES

APRIL 2024

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CONFIGURING THE ENVIRONMENT

This section shows how to create a virtual environment on **Linux** (though also tested on **MacOS**) and then install all the necessary dependencies. You are expected to have the **Python 3.11.3** interpreter ([Python Software Foundation official page](#)) and **Git** ([Project Git official page](#)) installed on your computer.

If you are going to configure your working environment on Ares clusters (PLGrid, Cyfronet infrastructure), you need to allocate resources and then load the appropriate Python module. The following command will allocate one CPU for 1 hour and then load the interpreter we need:

```
$ srun --time 00-01:00:00 --pty /bin/bash
$ module load python/3.11.3-gcccore-12.3.0
```

Let's now clone the repository, create a new virtual environment in it, activate the environment, update pip (package-management system) and install all the libraries stated in `requirements.txt` (available in the repository):

```
$ git clone https://github.com/klasocha/HiggsCP.git
$ cd HiggsCP
$ git checkout tyerniyazov-rhorho_refactored
$ python3.11 -m venv .venv
$ source .venv/bin/activate
(.venv) $ python -m pip install --no-cache-dir --upgrade pip
(.venv) $ pip install --no-cache-dir -r requirements.txt
```

The process of installing the libraries may take a while to finish. When it completes, your working environment is ready to run the code from the cloned repository. For Ares users, the first two steps in the violet box are regular. For all users, the step involving activating the virtual environment is regular. Don't forget to deactivate the environment if you finish experimenting with the project:

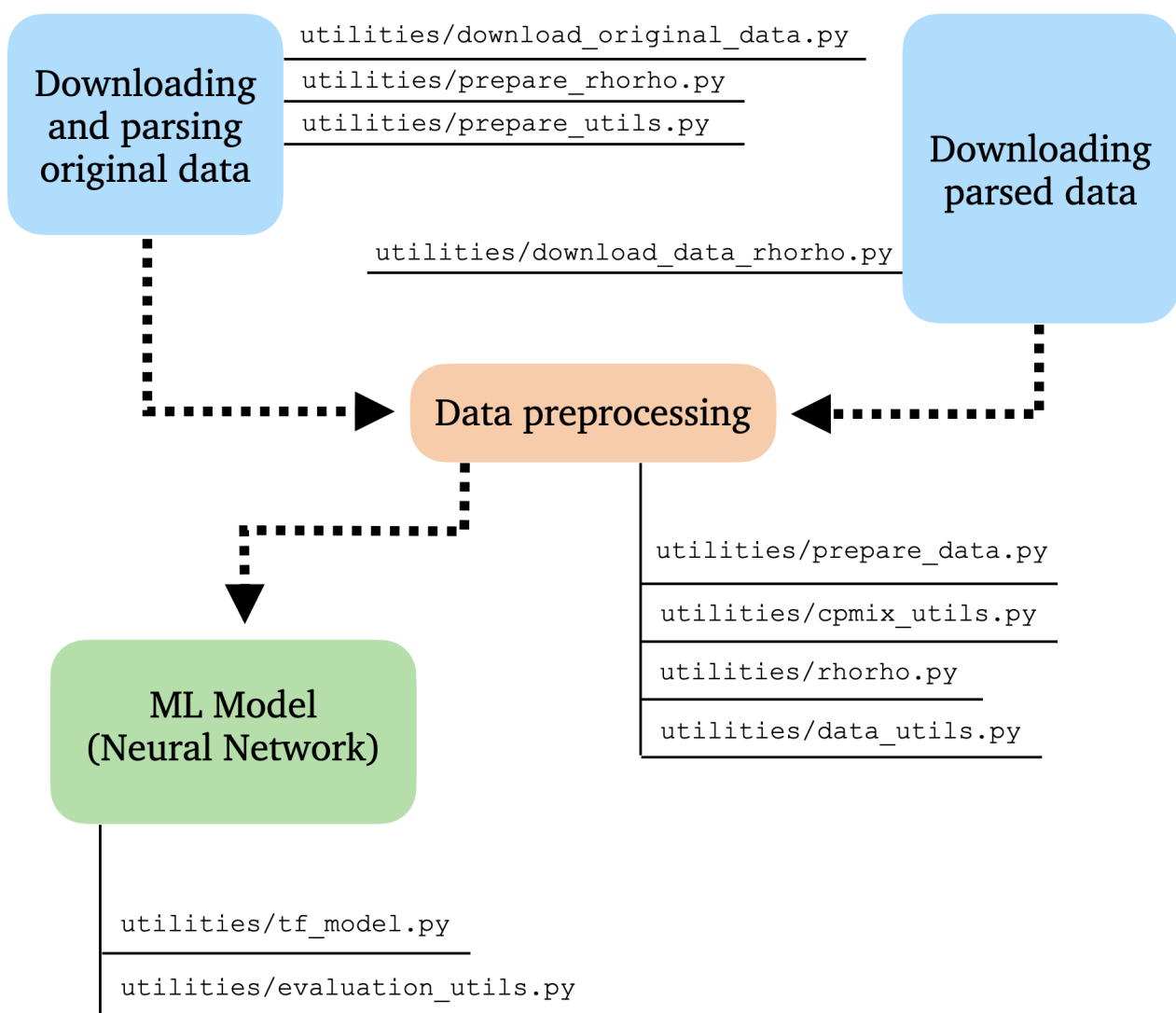
```
# Activating
$ source .venv/bin/activate

# Deactivating
(.venv) $ deactivate
```

ML FLOW

The whole project contains a number of packages responsible for implementing different parts of the machine learning flow. Each package contains several modules which are used by the main program `main.py` invoking particular actions, such as downloading data, training the model or even preparing some plots.

The below diagram depicts the names of some of the modules involved:



MODEL ARCHITECTURE

The ML model is represented by an artificial neural network which takes 6 different configurations. Depending on the type of configuration the architecture changes. Additionally, it is possible to specify some of the hyperparameters affecting the model structure and training process (optimizer, loss function, learning rate etc.)